

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP

ORGANIC CHEMISTRY: SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-09 CHEMISTRY

VISHAL SIR



JKBose Wallah

Topics To Covered

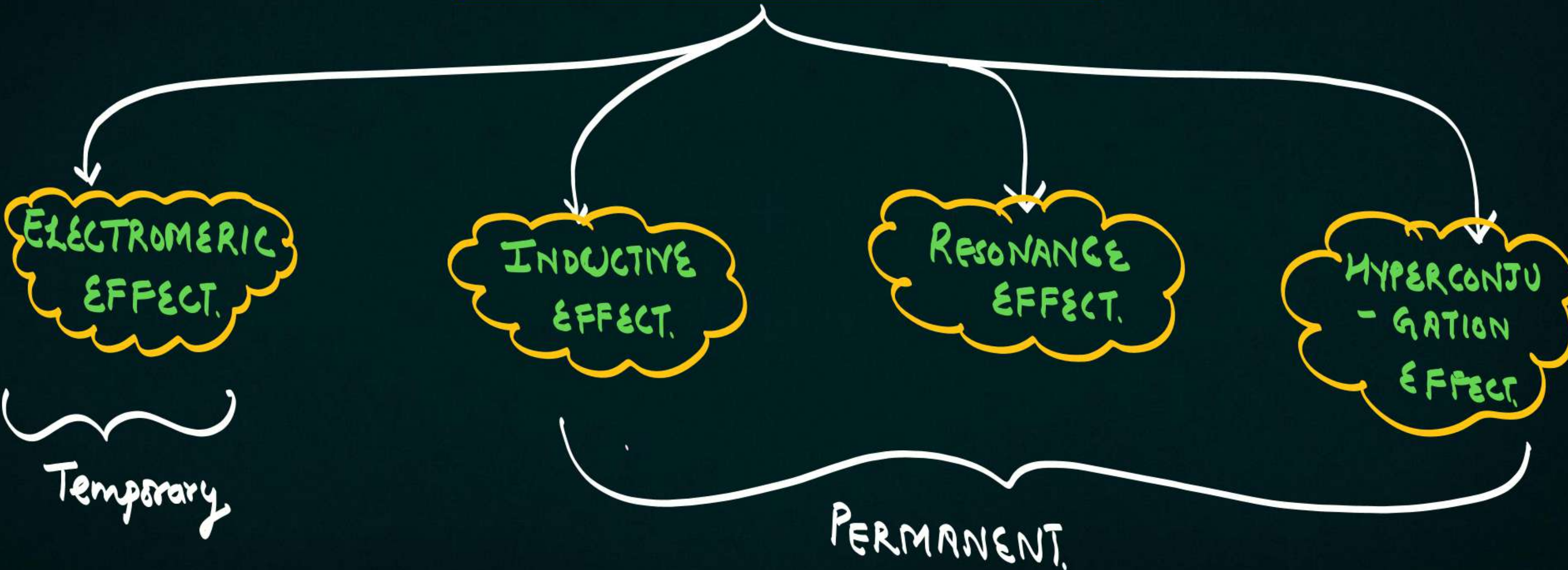
1 Electron Displacement Effects

NOMENCLATURE ✓

GOC ✓

Qualitative and Quantitative Analysis

ELECTRON DISPLACEMENT EFFECTS



ELECTRON DISPLACEMENT EFFECT:

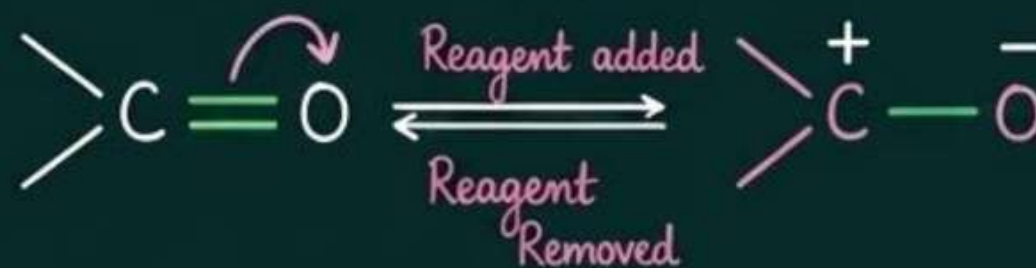
The displacement of bonding e^- pairs of covalent bond, within the molecule due to influence of external reagent or due to the some internal factors are called e^- displacement effect.

These are of four types.

- a) Electromeric effect b) Inductive effect
- c) Resonance d) Hyperconjugation

a) ELECTROMERIC EFFECT →

Complete transfer of e^- s of multiple bond (double or triple bond) to one of the bonded atoms (usually more electronegative atom), in presence of attacking reagent is called **electromeric effect**.



- It is **temporary effect**.

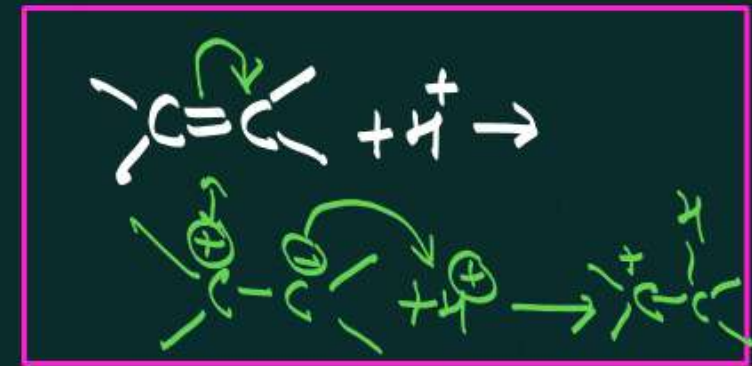
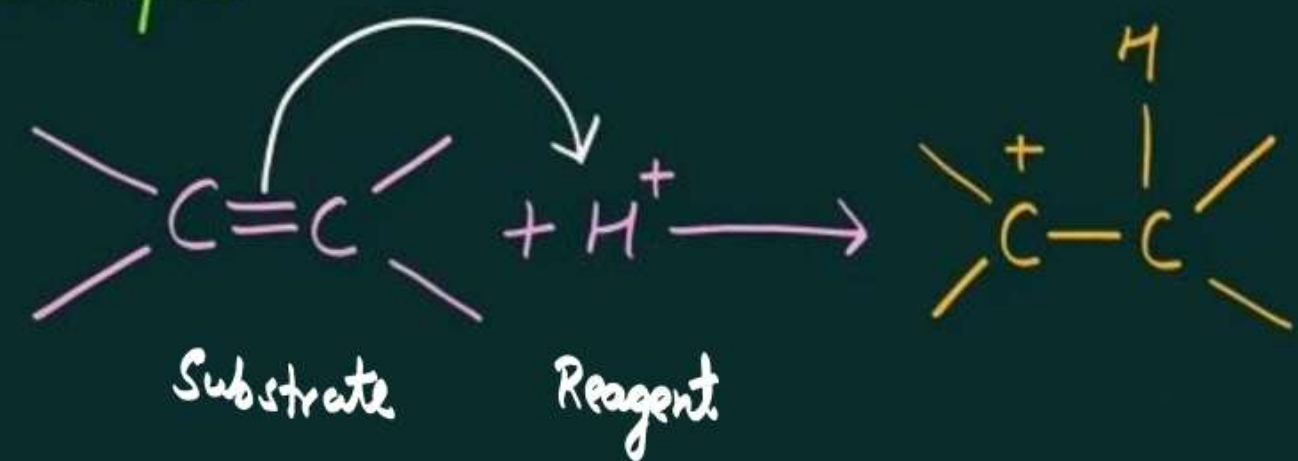
Electromeric effect is of two types.

- i) Positive Electromeric effect (+E).
- ii) Negative " " (-E).

Substrate + Attacking
Reagent

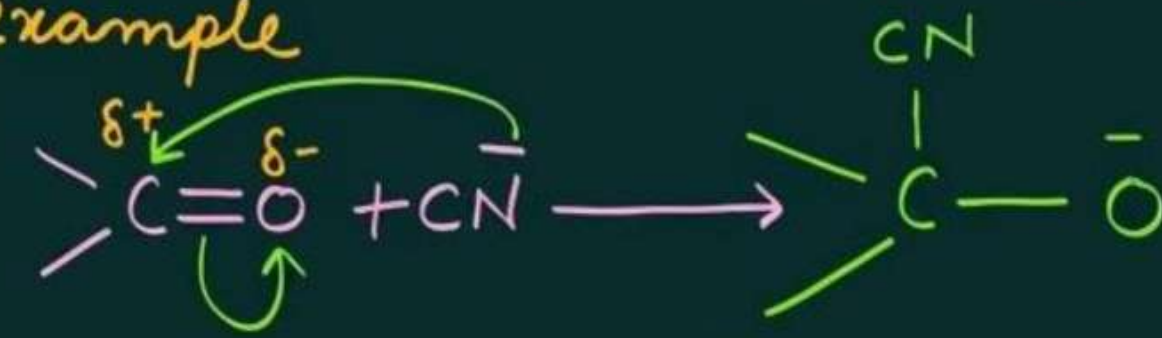
i) +E effect $\rightarrow$ If the electrons of π -bond are transferred to that atom to which reagents gets finally attached. (i.e towards the attacking reagent), the effect is called +E effect.

example

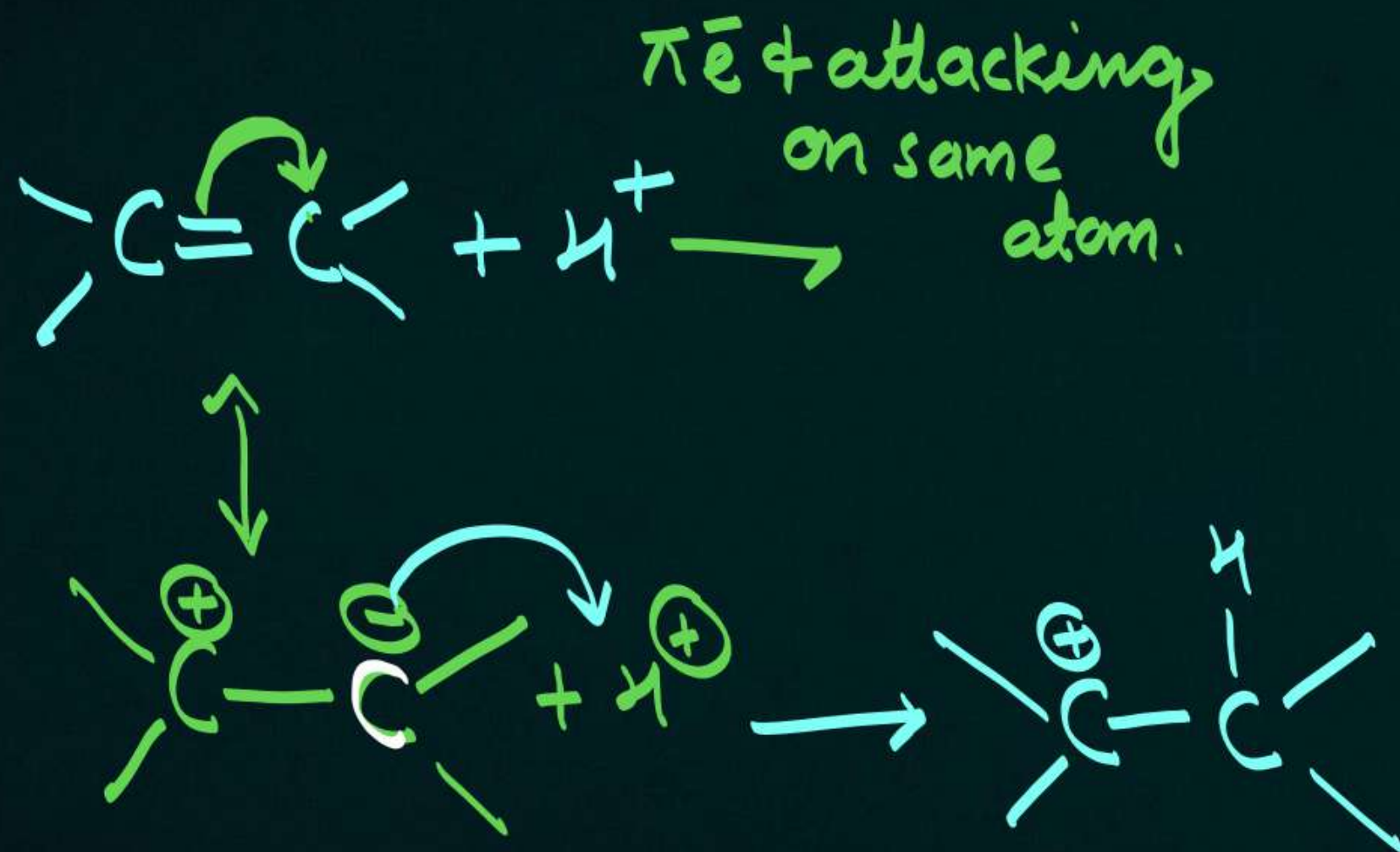


ii) -E effect $\rightarrow$ If electrons of the π -bond are transferred away from the attacking reagent, the effect is called -E effect.

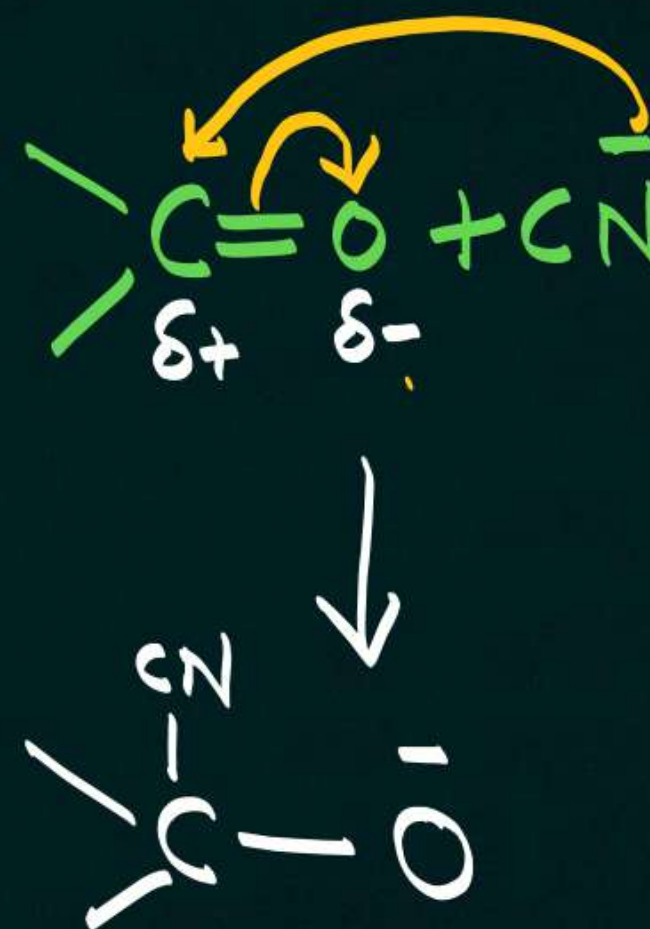
example



$+E$



$-E$



πe^- attacking reagent at different atom.

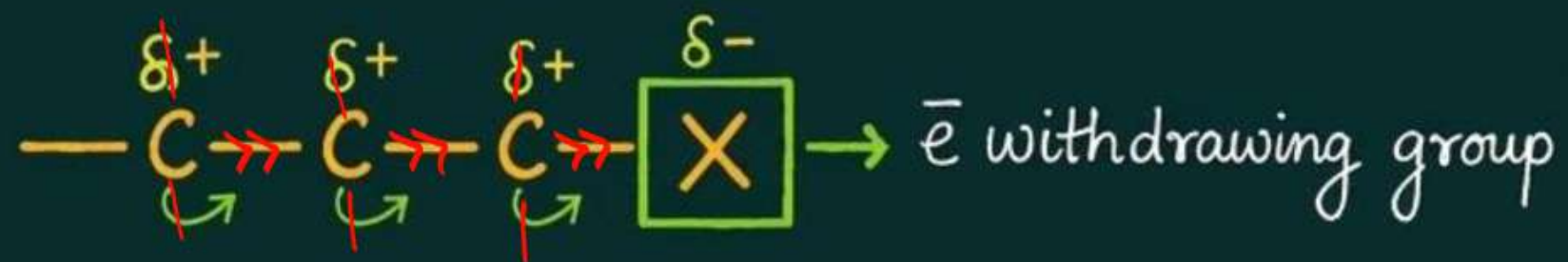
b) INDUCTIVE EFFECT → The displacement

of σ -electrons of a covalent bond within a saturated carbon chain whenever $\bar{e}$ withdrawing or electron donating group is attached at the end of the chain is called Inductive Effect (I-effect).

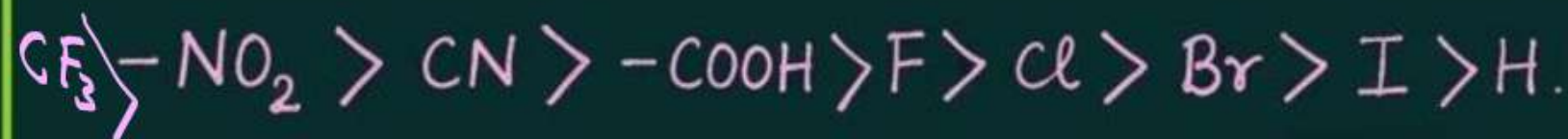
- This effect weakens steadily with increasing distance from the substituent (Σ DG or Σ WG).
- It becomes negligible after the 3-carbon atom.
- Two types i) +I ii) -I

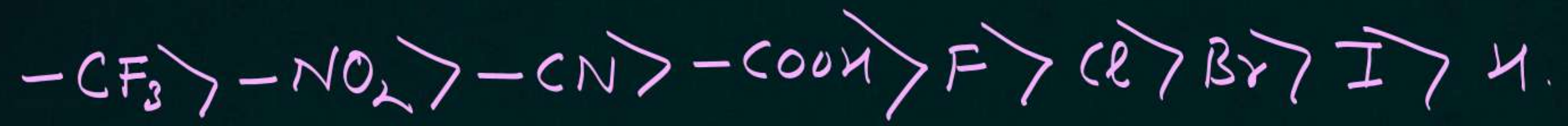
i) -I effect →

If the substituent attached on end of carbon chain is electron withdrawing, the effect is called -I effect. eg.

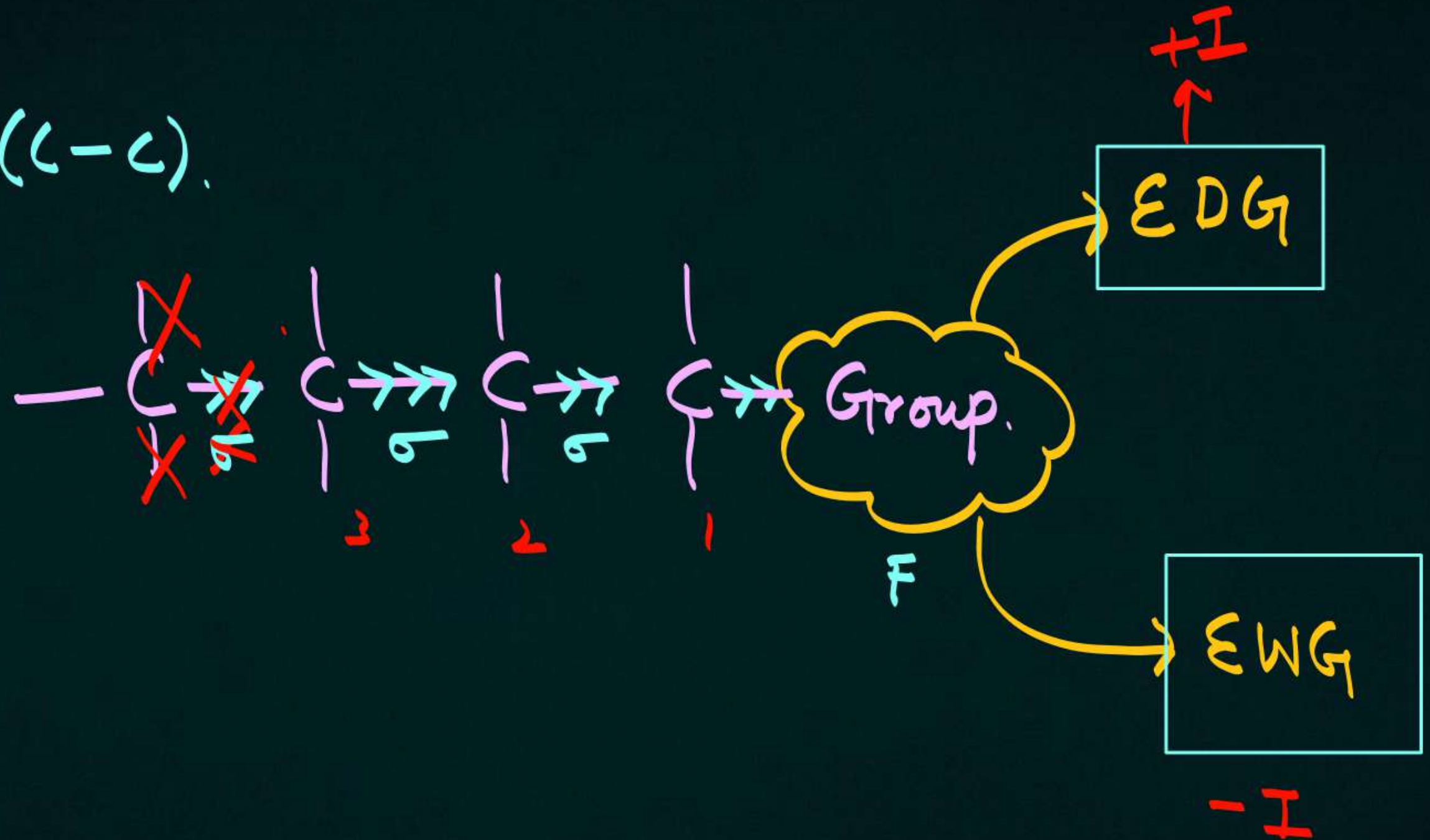


Order of -I effect with respect to hydrogen.





Saturated (C-C).

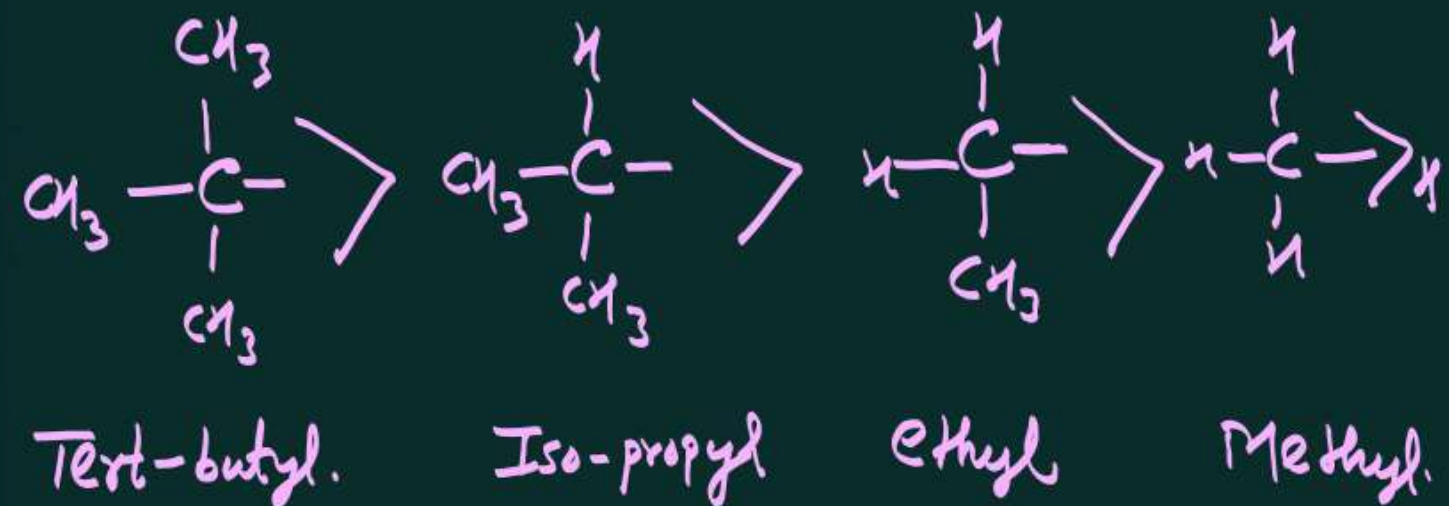
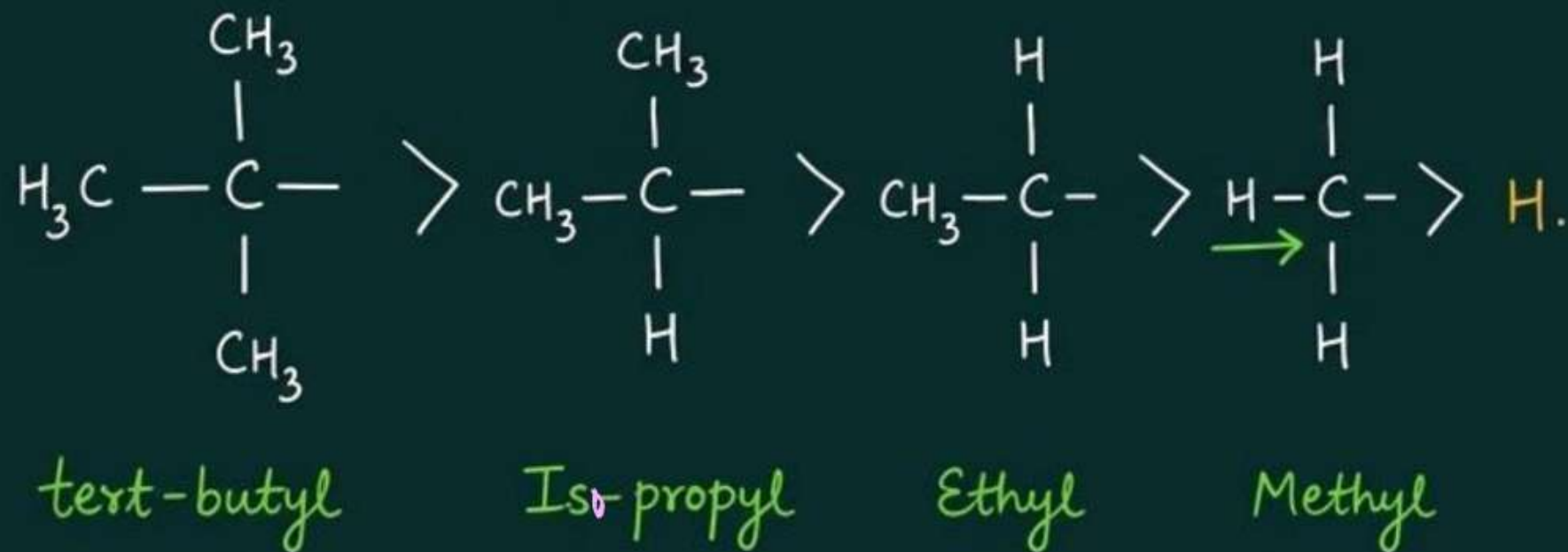


ii) +I effect → If the substituent present at the end of a carbon chain is electron donating, then effect is K/as +I effect.

e.g.



• Order of +I effect with respect to hydrogen.



★

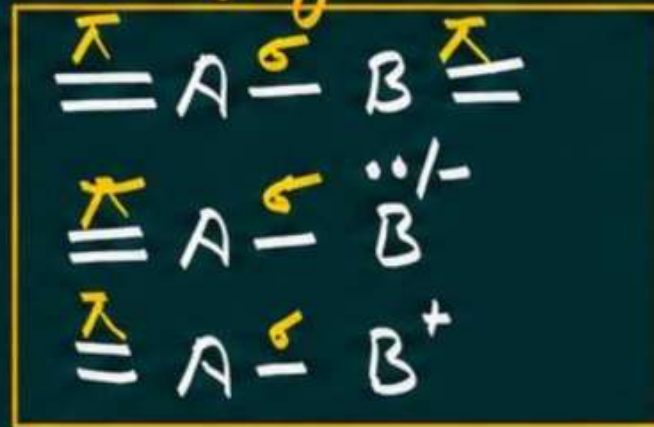


RESONANCE EFFECT → It is the polarity produced in a molecule by the interaction of two π bonds or one π bond and lone pair.

• This effect is produced in a conjugated systems.

• These are of two type
a) +R (positive Resonance) effect

b) -R effect (Negative Resonance).

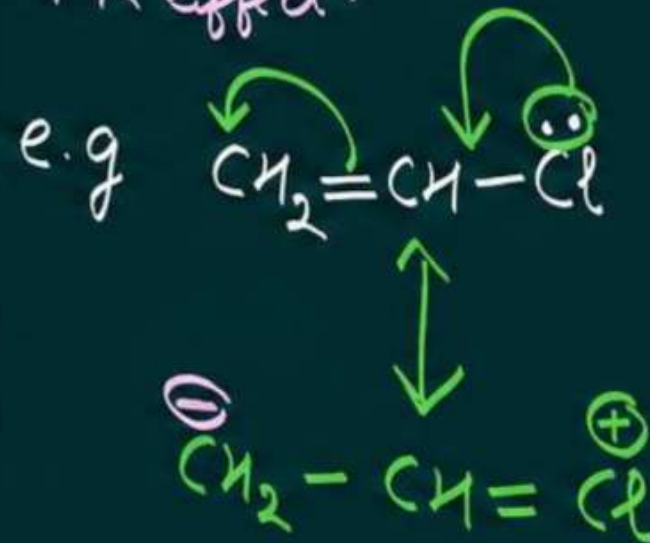


a) +R effect →

If the substituent has the tendency to donate electrons to the double bond or conjugated system, the effect is called positive resonance effect.

For example groups like.

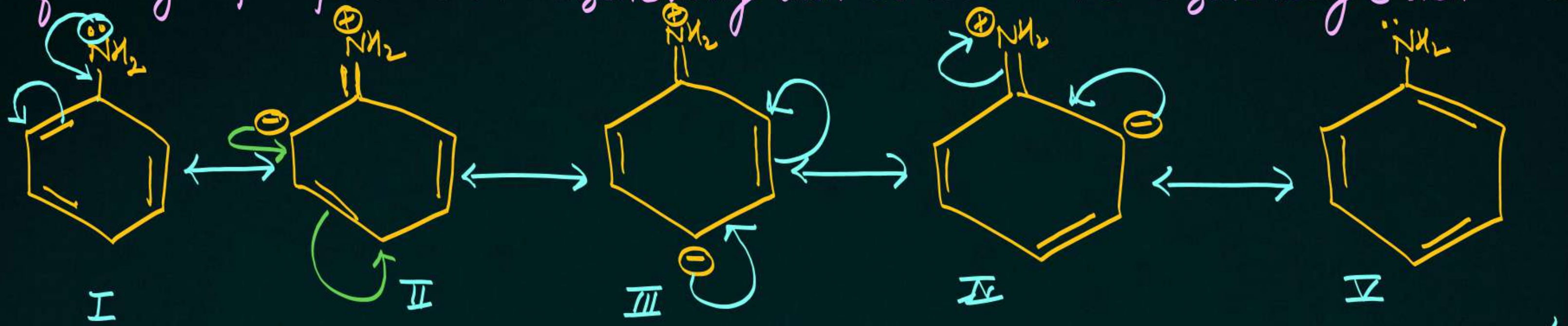
$-\ddot{O}H$, $-\ddot{N}H_2$, $-\ddot{N}R_2$, $-\ddot{Cl}$, $\ddot{Br}$, $\ddot{I}$ etc show +R effect.



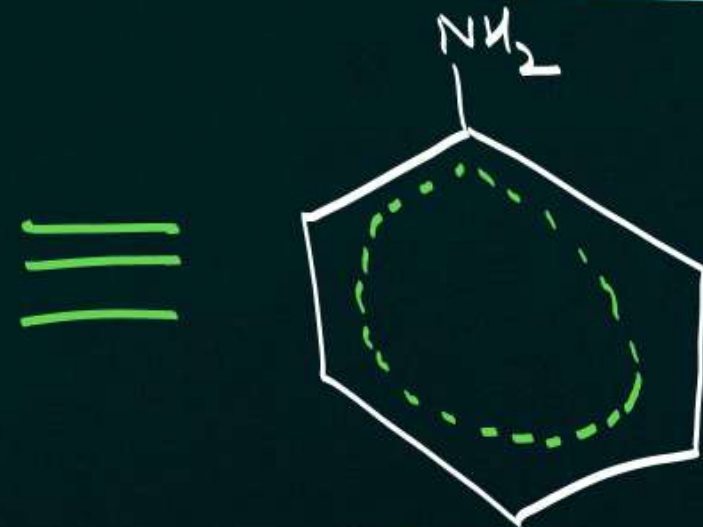
+R Group Ki Jaga pehchan,
Pehla hi atom par lone pair ka Nishan



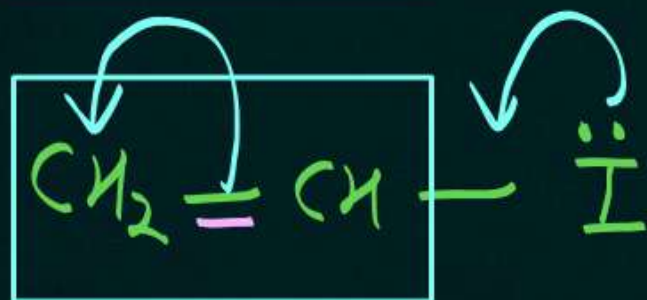
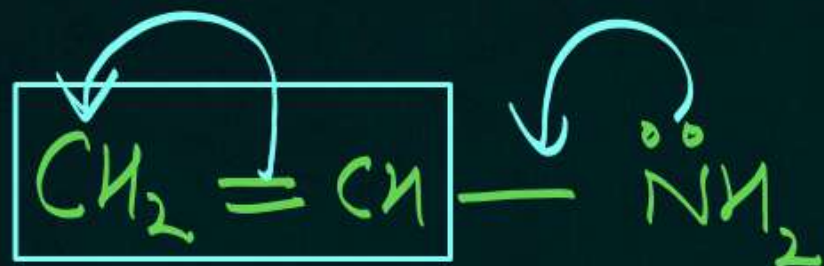
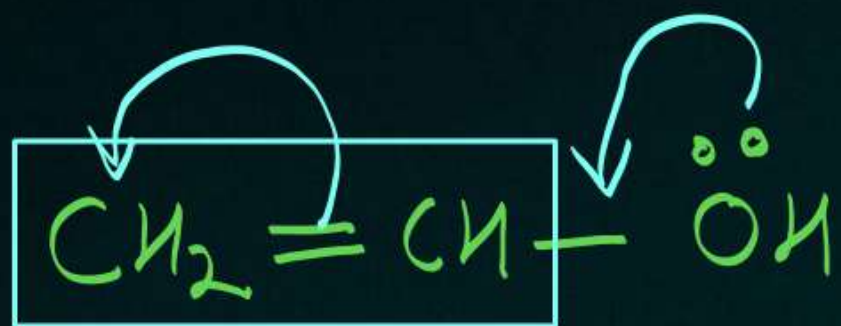
If +R group is present on benzene ring then it will make benzene ring $\bar{e}$ rich.



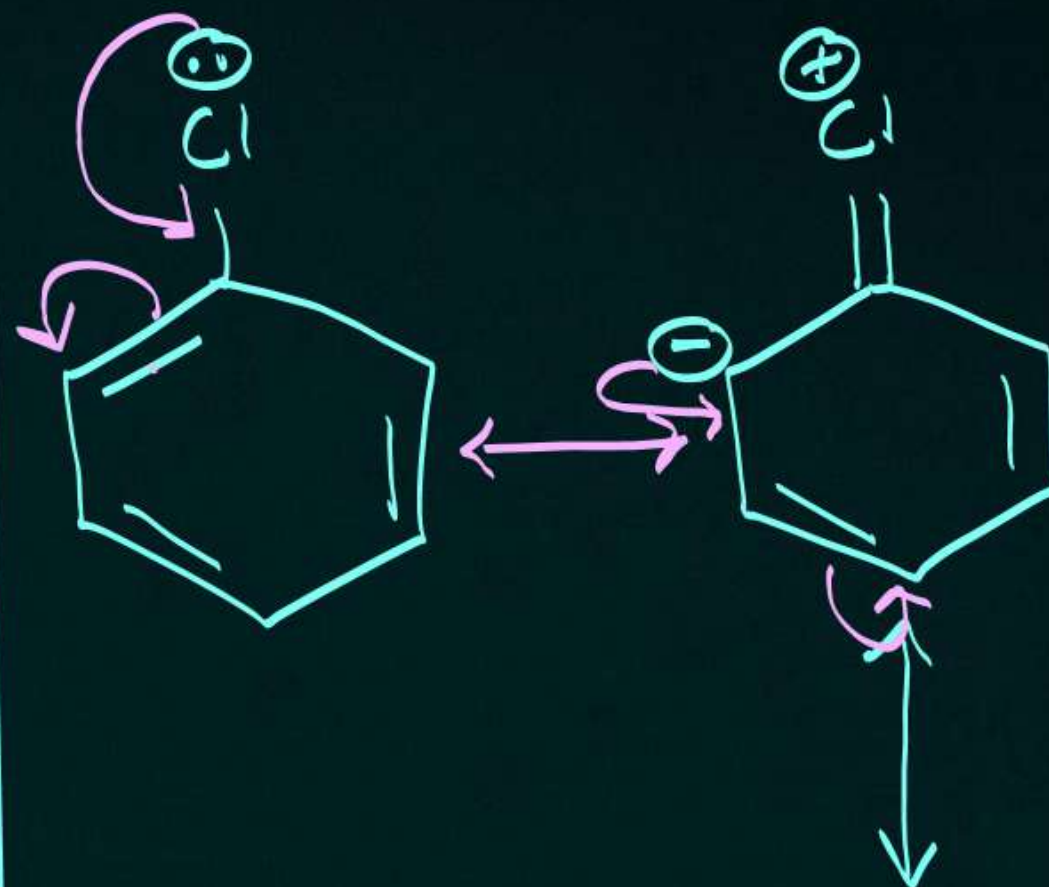
Resonating Structure.

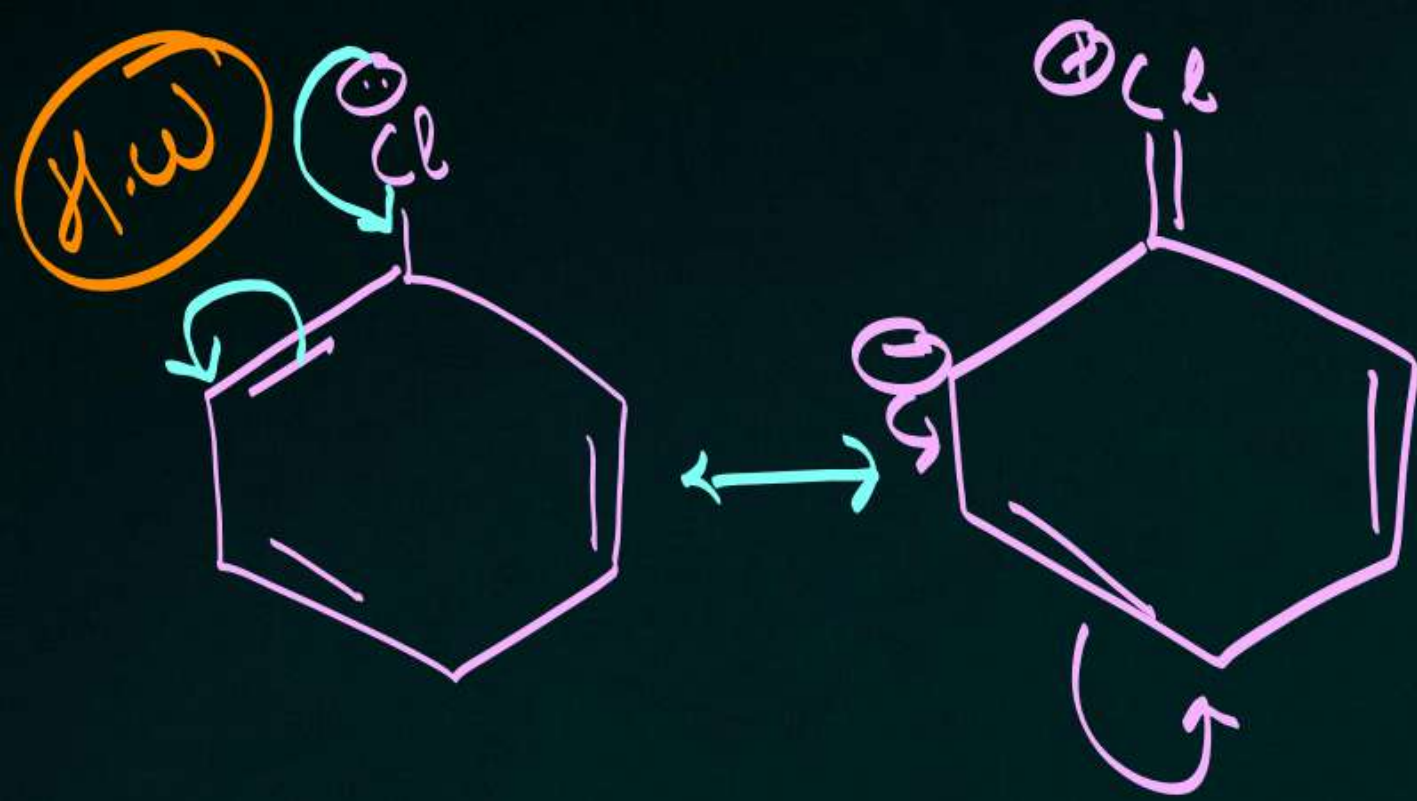


Resonance hybrid.



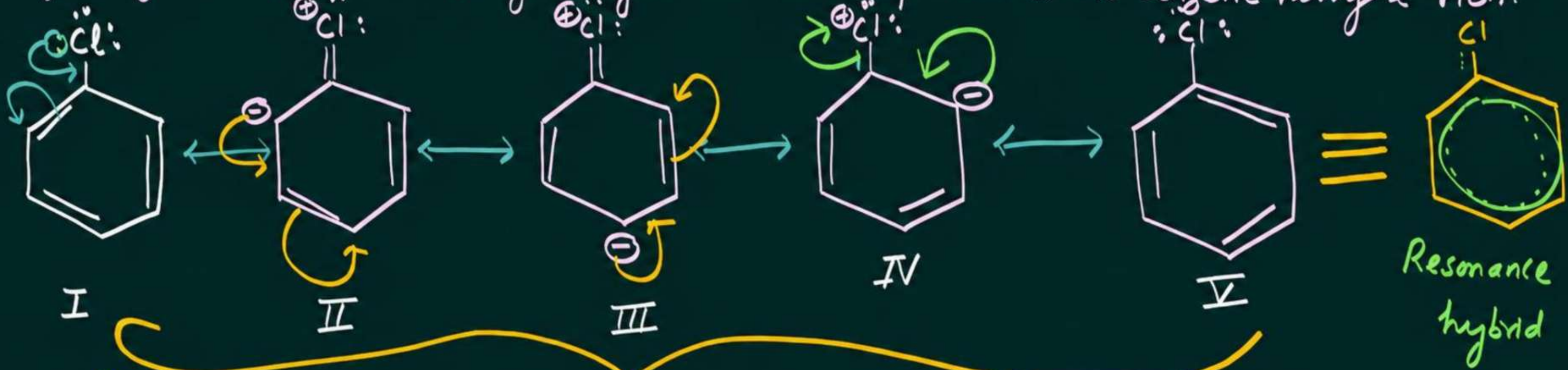
(+R)





PRACTICE

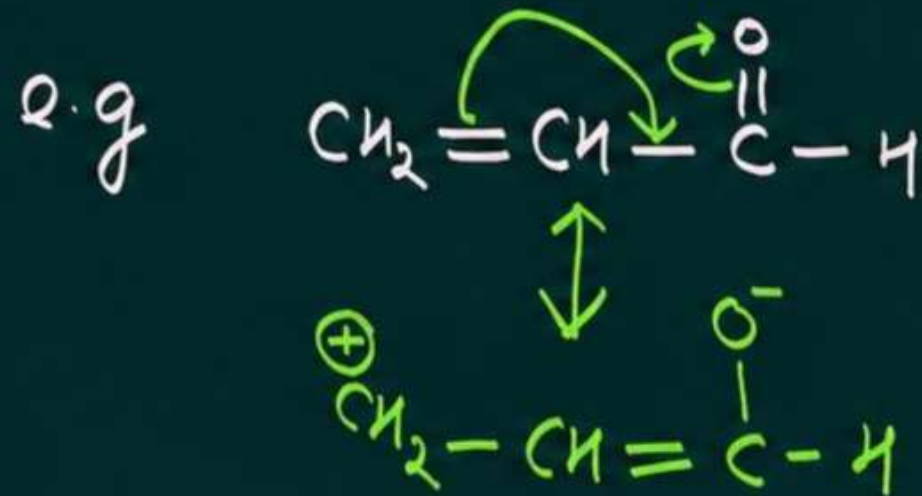
If +R group is present benzene ring, then +R Group makes the benzene ring $\bar{e}$ rich.



Due to +R effect $\bar{e}$ density increases on ortho & para position of benzene ring

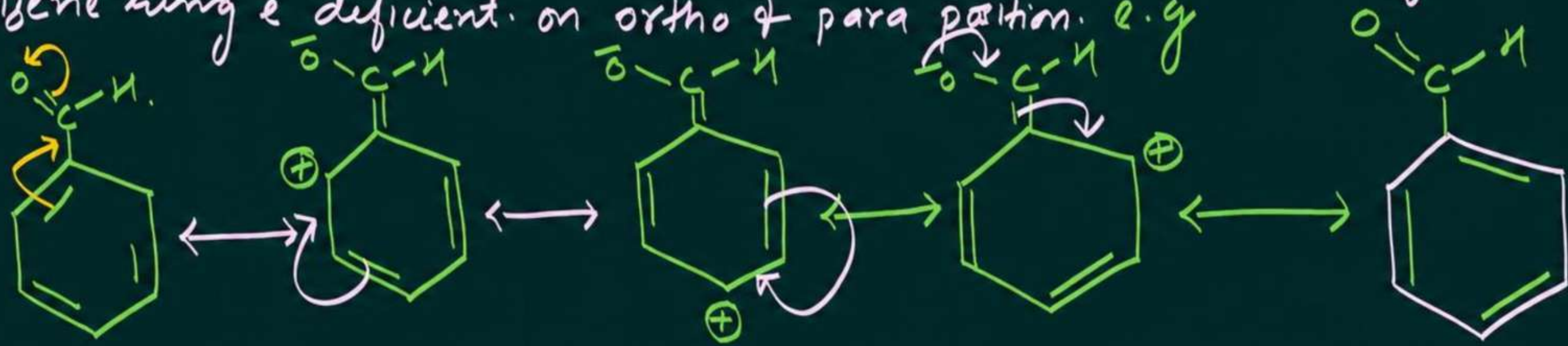
b) -R effect $\rightarrow$ If a substituent attached on double bond or a conjugated system has tendency to withdraw electrons from double bond to itself, the effect is called -R effect.

example. $\overset{\text{O}}{\parallel} \text{C}-\text{H}$, $\overset{\text{O}}{\parallel} \text{C}-$, $\overset{\text{O}}{\parallel} \text{C}-\text{OR}$, $\overset{\text{O}}{\parallel} \text{C}-\text{NH}_2$ etc. $\text{C}\equiv\text{N}$



-R-group ki
xya Pehchan
Pehla hi atom
pe double or MB
ka dishan.

If - R group is present ring, then it withdraw $\bar{e}$ from benzene ring & makes the benzene ring $\bar{e}$ deficient on ortho & para position. e.g.



4) Hyperconjugation $\rightarrow$

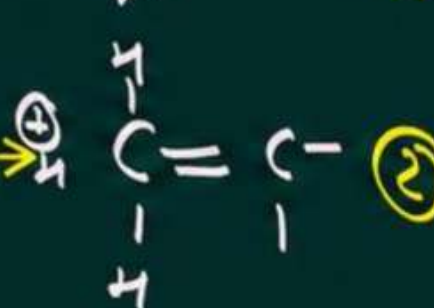
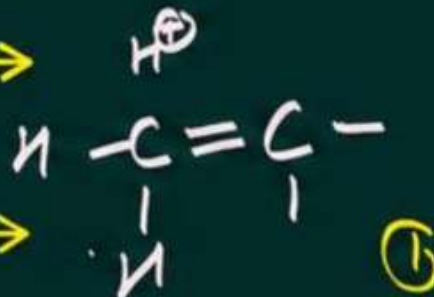
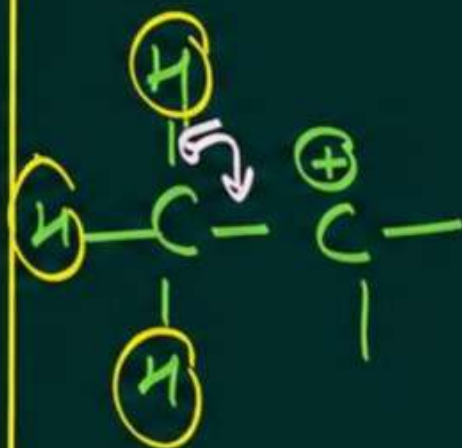
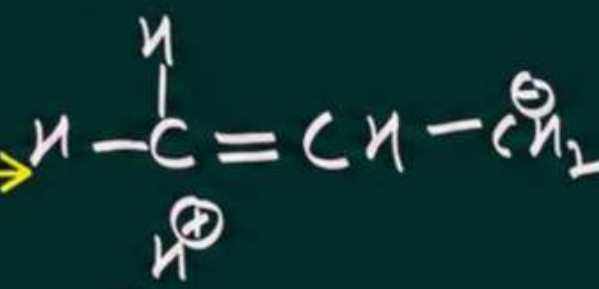
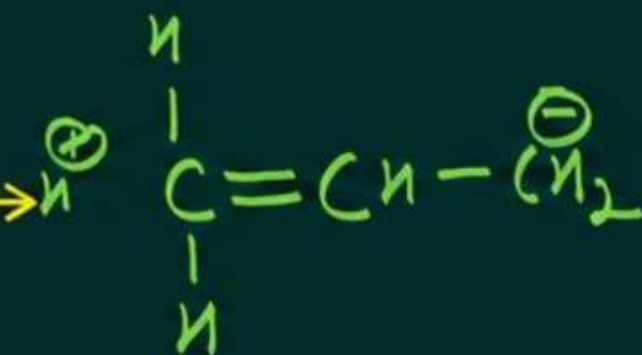
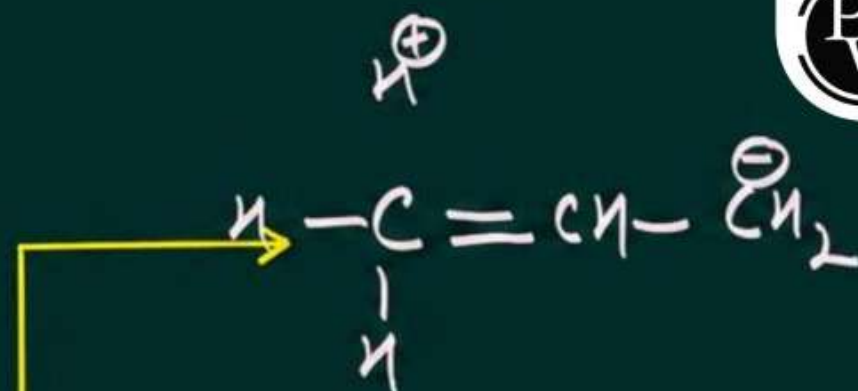
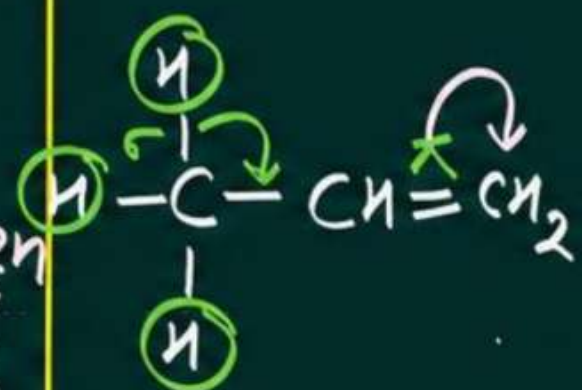
when
C-H bond is attached to an unsaturated system (double bond or benzene ring), the σ -e's of C-H bond interact with the unsaturated system. The interaction between e's of π -bond and σ electrons of C-H bond is called hyperconjugation.

This effect was Baker and Nathon i.e why

it is also known as Baker-Nathon effect

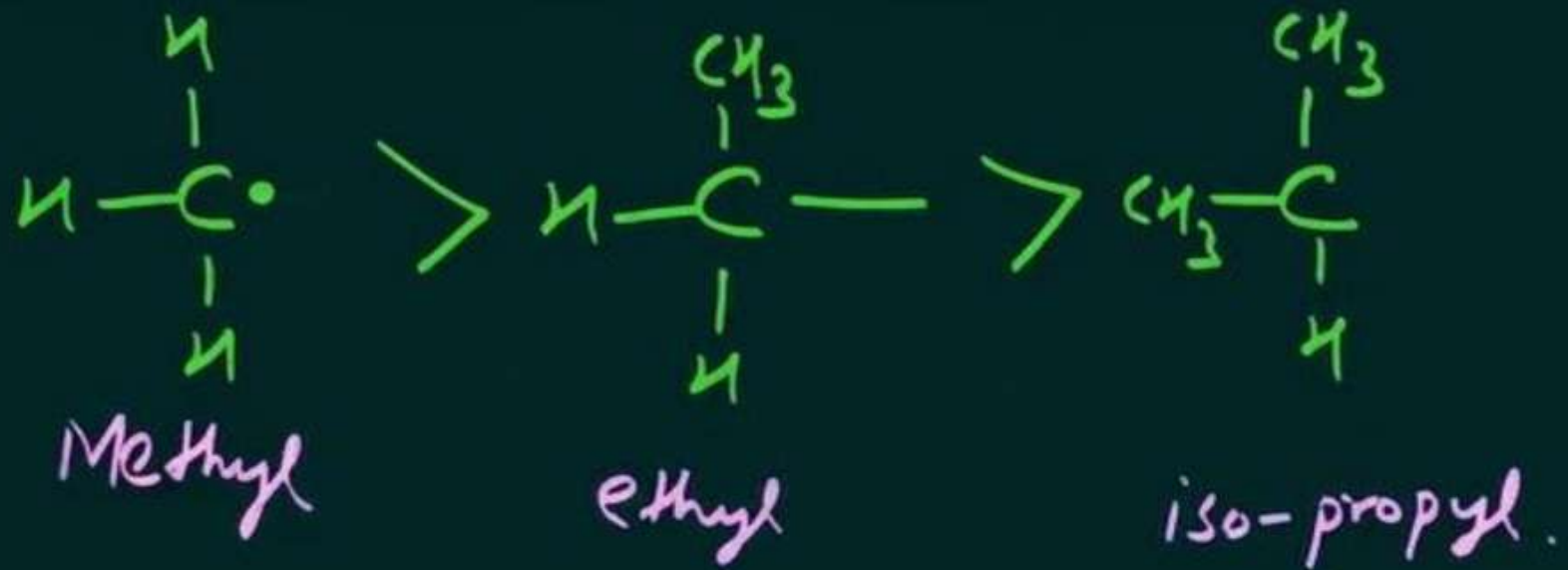
Also called No-bond Resonance.

e.g



(3) H.W

Order of hyperconjugation



PRACTICE KAR LENA.

Thank
You

#XUV 